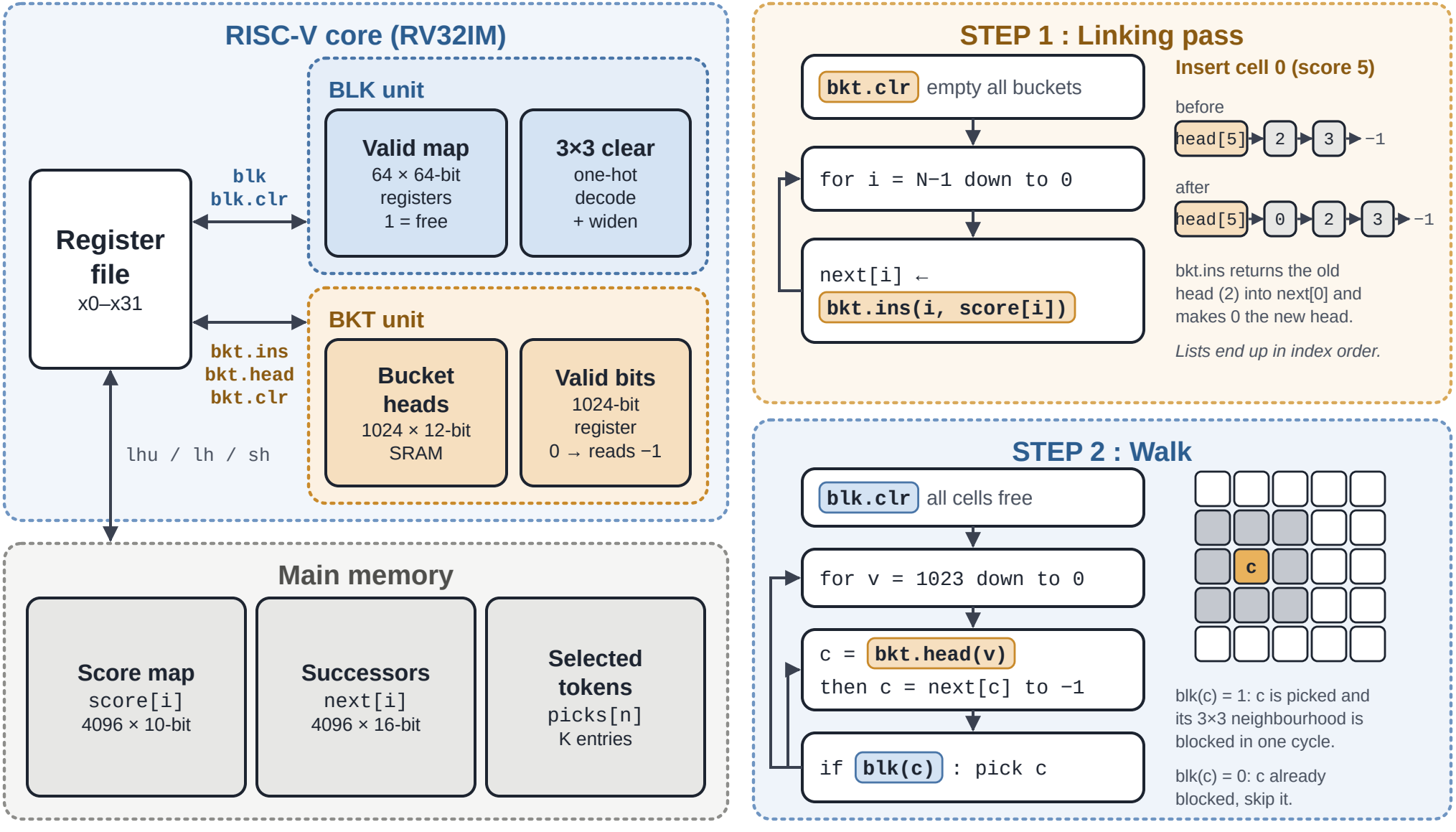

RISC-V core (RV32IM)
BLK unit
Valid map
64 × 64-bit
registers
1 = free
3×3 clear
one-hot
decode
+ widen
Register file
x0–x31
blk
blk.clr
BKT unit
Bucket heads
1024 × 12-bit
SRAM
Valid bits
1024-bit
register
0 → reads −1
bkt.ins
bkt.head
bkt.clr
lhu / lh / sh
Main memory
Score map
score[i]
4096 × 10-bit
Successors
next[i]
4096 × 16-bit
Selected tokens
picks[n]
K entries
STEP 1 : Linking pass
bkt.clr empty all buckets
for i = N−1 down to 0
next[i] ←
bkt.ins(i, score[i])
Insert cell 0 (score 5)
before
head[5] 2 3 −1
after
head[5] 0 2 3 −1
bkt.ins returns the old head (2) into next[0] and makes 0 the new head.
Lists end up in index order.
STEP 2 : Walk
blk.clr all cells free
for v = 1023 down to 0
c = bkt.head(v)
then c = next[c] to −1
if blk(c) : pick c
c
blk(c) = 1: c is picked and its 3×3 neighbourhood is blocked in one cycle.
blk(c) = 0: c already blocked, skip it.